

TAP AGAINST ME

GAME DESIGN BIBLE

Every Tap Against Me Takes You Further.

Comprehensive Product, Gameplay, Systems, Content, and Technical Vision

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Don't Watch Media

Document Control and Canon Framework

PURPOSE This bible consolidates the full Tap Against Me vision discussed to date. It is intended to be the central reference for design, engineering, art, audio, economy, live operations, testing, and roadmap decisions.

Tap Against Me has evolved quickly. Several ideas were explored before the game settled into its current identity as an airborne tapping-survival game with competitive play and an expansive, noncompetitive Infinity Taps adventure. This document preserves that history while separating current canon from optional, future, and legacy concepts.

Status vocabulary

Label	Meaning
CURRENT CANON	A committed direction that should guide current implementation unless explicitly changed.
MVP	Required for the first credible playable or first public release, depending on the section.
POST-MVP	Approved direction, but not required to prove the core product.
LONG-TERM VISION	The intended mature form of the system after the foundation is stable.
PROPOSED / TUNABLE	A concrete starting value or design option that must be validated through playtesting and analytics.
LEGACY / RECONCILE	An earlier concept that may contain useful ideas but conflicts with or predates current canon.

Core non-negotiables

- **Immediate readability.** A new player must understand within seconds that tapping keeps the character airborne and touching the ground ends a competitive run.
- **High input satisfaction.** Every valid tap must feel responsive through movement, animation, particles, audio, and optional haptics.
- **No pay-to-win.** Money may alter appearance, presentation, collection, and ad exposure, but never competitive power, physics, drop odds, ranking, or progression speed in a way that creates gameplay advantage.
- **Meaningful solo play.** The game must remain enjoyable when no human opponent is online.
- **Fair competition.** Human and bot matchmaking must produce opponents who feel challenging but realistically beatable.
- **Distinct emotional modes.** Competitive TAM is tense and skill-focused; Infinity Taps is playful, exploratory, reactive, and pressure-free.
- **Permanent identity.** Characters, cosmetics, badges, achievements, records, and discoveries should create attachment across short sessions and long-term play.

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1. Executive Vision

Tap Against Me is an Android-first, portrait-oriented 2D tapping game built around one immediately understandable physical rule: the player falls, tapping creates upward movement, and contact with the ground ends a competitive run. That simple rule supports short solo sessions, local and online competition, ranked progression, cosmetic identity, permanent achievements, and a radically different pressure-free exploration experience called Infinity Taps.

PRIMARY PROMISE Tap Against Me turns a single repeated input into mastery, rivalry, discovery, self-expression, and long-term progression.

The three layers of the product

Layer	Player promise	Primary systems
Immediate arcade loop	Tap now, recover, survive, and improve the next second.	Gravity, directional movement, pickups, hazards, close calls, responsive feedback.
Competitive identity	Prove skill against players or believable bots and climb a seasonal ladder.	Online/private/local play, Glicko-2, tiers, matchmaking, bots, leaderboards, seasons.
Persistent universe	Build a personal identity and discover reasons to return beyond winning.	Heroes, variants, cosmetics, badges, achievements, Infinity Taps, seasonal content.

The meaning of “Me”

The title deliberately avoids assigning “Me” to one literal mascot. Its meaning changes with context:

- **In solo survival**, “Me” is gravity, fatigue, the player’s previous best, and the moment they normally lose control.
- **In multiplayer**, “Me” is the opponent sharing the sky.
- **In ranked play**, “Me” is the ladder, the player’s current limit, and the pressure to perform consistently.
- **In Infinity Taps**, “Me” is distance, curiosity, the world itself, and the player’s willingness to continue exploring.

The title therefore supports both direct rivalry and self-improvement. The game should not introduce a canonical character literally named “Me,” because doing so would narrow the title’s symbolic strength.

Brand language

Use	Line
Official slogan	Every Tap Against Me Takes You Further.
Short supporting line	Tap. Rise. Explore.
Competitive positioning	Tap to survive. Tap against friends. Tap against the world.
Design principle	The next tap should always matter.

2. Product Identity and Positioning

Genre and identity

The current game is best described as a portrait mobile airborne survival game with competitive tapping, persistent cosmetic progression, and an optional infinite exploration adventure. It is not an idle clicker, despite tracking large

tap totals. It is not a traditional platformer, despite directional movement. It is not solely a multiplayer game, because solo survival, bots, local play, and Infinity Taps must remain independently valuable.

Design pillars

Pillar	Interpretation	Failure to avoid
One-touch depth	The input is simple, but timing, placement, momentum, pickups, risk, and pressure create mastery.	Adding unnecessary buttons or systems that make tapping secondary.
Readable intensity	Difficulty rises clearly through gravity, movement demands, hazards, and opponent pressure.	Random deaths that the player cannot understand or learn from.
Expressive identity	Heroes, skins, particles, tap effects, badges, frames, and animations make players recognizable.	Cosmetics that obscure collision, items, hazards, or competitive clarity.
Fair rivalry	The best opponent is one the player can beat with a slightly stronger-than-normal run.	Extreme mismatches, cheating bots, hidden advantages, or purchased power.
Living discovery	Infinity Taps makes taps affect the environment, reveal secrets, and produce delight.	A decorative endless background that ignores player input.
Respectful monetization	Spending supports expression and convenience without manipulating competition.	Permanent power boosts, ranked advantages, or interruptive live-round ads.

Competitive TAM and Infinity Taps

The project contains two experiences with different emotional goals. This is intentional, but the distinction must remain clear in product design and production scope.

Dimension	Competitive TAM	Infinity Taps
Emotion	Tense, focused, triumphant, nerve-driven.	Relaxed, curious, playful, satisfying.
Session	Short rounds and repeat matches.	Open-ended fidgeting and exploration.
Failure	Ground contact or mode-specific defeat ends the round.	No traditional loss state.
Progress	Wins, rating, rank, survival performance, seasonal rewards.	Tap total, distance, biomes, discoveries, encounters, exclusive unlocks.
Content pressure	Requires tight mechanics, balance, networking, and fair opponents.	Requires reactive environments, biomes, secrets, creatures, music, and discovery density.
Population dependency	Mitigated through solo play, bots, ghosts, and private play.	None for core enjoyment.

PRODUCT STRUCTURE Current direction treats Infinity Taps as a major mode inside the TAM ecosystem, while preserving the option to separate it into a sister game later if its content scope and audience become substantially independent.

3. Audience, Platform, and Session Design

Platform priorities

- **Primary platform:** Android smartphones.
- **Primary orientation:** Vertical/portrait, designed for one-handed or two-thumb mobile use.
- **Secondary input:** Optional controller support may be retained, but it must not create competitive input advantage or distract from touch-first design.
- **Future portability:** The architecture may support additional mobile or desktop builds, but the interface and performance budgets must be established from the Android-first baseline.

Player motivations

Player type	What they seek	Systems that serve them
Quick-session player	A complete satisfying experience in under several minutes.	Solo run, casual match, instant rematch, short challenge.
Skill improver	Evidence that timing, recovery, and decision-making are improving.	Personal bests, survival averages, match history, rank climb, performance stats.
Competitor	Fair opponents, visible status, seasonal stakes, recognition.	Ranked ladder, Glicko matchmaking, leaderboards, divisions, seasonal rewards.
Collector	Characters, variants, badges, effects, and rare unlocks.	Shop, achievements, seasonal pass, Infinity exclusives, discovery journal.
Fidget/relaxation player	Pressure-free tapping with pleasing audiovisual feedback.	Infinity Taps, no-loss exploration, reactive world, ambient music.
Social player	Friends, private matches, comparisons, shared milestones.	Profiles, friend search, private rooms, friend leaderboards, events.

Target session patterns

- **Micro-session:** One run, one daily challenge check, one reward claim, or a short Infinity Taps interaction.
- **Standard session:** Several solo or ranked matches, loadout changes, challenge progress, and a rematch loop.
- **Extended session:** Rank pushing, friend competition, long Infinity exploration, seasonal objectives, or discovery hunting.

4. Core Competitive Gameplay

Foundational rule

CURRENT CANON The character begins by falling from the sky. Each valid tap creates upward movement or an impulse toward the tapped area. Gravity continuously pulls the character downward. Touching the ground ends the competitive run.

Moment-to-moment loop

- Read the character's current velocity, altitude, and trajectory.
- Choose where and when to tap rather than tapping without thought.
- Convert the tap into upward or directional movement.
- Correct oversteer, understeer, and momentum before the next danger window.
- Decide whether an item is worth leaving a safe path to collect.
- Respond to escalating gravity and arena conditions.
- Use close calls and recoveries to create dramatic survival moments.
- Outlast the opponent, meet the current mode objective, or extend a personal record.

Directional tapping

The input should preserve immediate accessibility while allowing spatial skill. A tap is not merely a generic jump button; its screen position should influence the direction of the impulse. Tapping left of the character should bias movement left and upward, tapping right should bias right and upward, and tapping above or near the character should create a more vertical correction. Exact impulse math is tunable, but it must be predictable enough for mastery.

Property	Design requirement
Responsiveness	The movement response begins immediately on valid input; audiovisual effects must not delay physics.
Predictability	Similar taps under similar movement states should produce similar results.

Property	Design requirement
Momentum	Movement should preserve enough inertia to make recovery and planning meaningful.
Skill ceiling	Expert players should distinguish between panic tapping, efficient stabilization, item pursuit, and intentional positioning.
Readability	The player should understand why a tap moved the character in a particular direction.

Escalating gravity

Gravity increases over time, ensuring that survival cannot remain indefinitely static. The escalation should be gradual, telegraphed, and tuned to create distinct performance bands. Early play teaches control; mid-run play tests consistency and pickup judgment; late-run play demands fast recovery and efficient tapping. The curve should create meaningful differences between a 15-second beginner, a 63-second developing expert, and a 72-second higher-skilled player.

- **Do not create invisible spikes.** Changes should be smooth or announced through audio, visual atmosphere, UI, or predictable phase transitions.
- **Do not reward pure tap spam by default.** Uncontrolled tapping should create inefficient movement, overshoot, or poor positioning.
- **Do not make character cosmetics alter the curve.** Physics must remain standardized in ranked play.

Loss and result clarity

Ground contact is a decisive state. The game should capture the exact contact moment, resolve simultaneous or near-simultaneous outcomes consistently, and show the player how close the result was. Result screens should prioritize survival time, opponent comparison, personal-best status, rating change, challenge progress, and earned rewards.

5. Pickups, Hazards, and Match Dynamics

Established pickups

Pickup	Current function	Design purpose
Green Apple	Temporarily increases jump force or upward impulse.	Creates recovery opportunities, route choices, and aggressive movement possibilities.
Purple Fruit / Poison Fruit	Increases gravity or applies a gravity-related debuff.	Introduces risk, disruption, achievement goals, and survival-under-pressure moments.
Golden Hero Token	A valuable currency pickup used toward the shop/economy.	Adds optional risk and collection incentive during competitive or solo play.

Pickup design rules

- **Items must create decisions.** A pickup positioned directly on the safest path is a reward; one placed off-path asks the player to trade safety for value.
- **Spawn fairness matters.** Ranked items must not create impossible, opponent-exclusive advantages because of camera, latency, or spawn asymmetry.
- **Effects need clear telegraphing.** Players must recognize item type, duration, and active state without reading text during a match.
- **Bots follow perception rules.** They must notice, evaluate, and pursue items with human-like delay and imperfect judgment.

- **Achievements may encourage risky behavior, but not sabotage normal competition.** Examples involving poison consumption or greed should remain optional goals.

Item response and collection feel

Collection should produce a concise burst of feedback: pickup animation, sound, small UI confirmation, and a visible state change. The effect should be exciting without obscuring the character or opponent. For Golden Hero Tokens, the value should be legible, but the game must avoid making players feel that losing a match was irrationally punished because they skipped a token.

Future item framework

Future pickups should fit one of four functions: survival aid, movement modifier, opponent pressure, or optional value. Every new item must pass competitive fairness review, bot-behavior review, visual clarity review, and no-pay-to-win review before ranked inclusion.

6. Game Modes

Core launch modes

Mode	Objective	Population requirement	Launch role
Solo Survival	Remain airborne as long as possible and improve personal performance.	None	Primary practice, onboarding, and always-available core loop.
Local Competitive	Compete on one device or supported local setup under shared rules.	Nearby players	Immediate social play without backend dependence.
Private Online Match	Invite or find a known player and compete directly.	Two players	Friend play and dependable testing.
Casual Matchmaking	Find a human near skill or receive bot backfill.	One player minimum	Low-friction competitive play.
Ranked Matchmaking	Compete for seasonal rating, divisions, and rewards.	One player minimum at lower tiers due to bots	Long-term competitive progression.
Infinity Taps	Explore, fidget, interact, discover, and grow permanent tap count without losing.	None	Relaxation, universe-building, collection, and content depth.

Competitive mode concepts

Mode concept	Rule
Survival / Last Airborne	The player who remains airborne longest wins. This is the cleanest expression of the core mechanic.
First to Zero	A mode where a defined resource, survival meter, or opponent state reaches zero first. The exact resource model remains to be finalized.
Most Taps	Score is determined by valid taps under a time or survival constraint. Anti-macro and efficiency rules are essential.
Dual Victory	A hybrid condition combining survival with another measurable objective, such as score, item control, or tap target.
Free-for-All	More than two players share a competitive space; last airborne or highest qualified performance wins.
Tournament / Elimination	Bracketed or staged competition for live events and future community play.

SCOPE NOTE Survival / Last Airborne is the primary canonical competitive rule. Other named modes are approved concepts, but their exact rules should not expand the MVP until the core match is polished.

Asynchronous modes

- **Ghost Challenge.** Replay a recorded player performance transparently as a ghost rather than pretending it is live.
- **Daily Survival Seed.** All players face the same arena/item sequence and compare results.
- **Friend Benchmark.** Challenge a friend's recorded time or movement sequence.
- **Community Milestone.** Aggregate taps, survival, or discoveries toward a shared goal.

7. Competitive Match Flow and User Experience

Recommended match flow

- Player selects Solo, Casual, Ranked, Private, or Local.
- Loadout is validated server-side for owned cosmetics and legal competitive configuration.
- Matchmaking searches for an appropriate human and expands within defined limits.
- A clearly identified bot is selected when policy permits and no appropriate human is available.
- Opponent introduction shows username, hero, primary badge, visible rank, and selected presentation cosmetics.
- Countdown begins; current backend configuration has used a 3,000 ms countdown as an implementation baseline.
- Server-authoritative match simulation validates inputs, state, items, survival, and result.
- Result screen shows winner, both survival times, close finish, rating/rank change, rewards, challenge progress, and rematch options.

Match introduction

The introduction should be short enough not to slow repeat play, but valuable enough to make profiles and cosmetics matter. One primary badge should appear beside the player name. Up to three showcase badges remain visible on the full profile rather than crowding the match intro.

Result screen priorities

Priority	Information
1	Victory/defeat/draw result and exact survival times.
2	Personal best, seasonal best, or meaningful improvement callout.
3	Glicko rating change and visible rank-star/division progress.
4	Rewards, achievement progress, and daily/weekly challenge progress.
5	Rematch, continue queue, change loadout, view opponent profile, or report.

Close finishes

Photo-finish outcomes should be celebrated because they reinforce fairness and motivate rematches. A dedicated “Photo Finish” badge or achievement already fits the broader badge vocabulary. Resolution must be based on authoritative timestamps, not client visual order.

8. Skill Measurement and Glicko-2 Ranking

Rating model

CURRENT CANON Use Glicko-2 as the hidden competitive skill model. Visible rank tiers provide understandable progression, while Glicko rating, deviation, and volatility drive matchmaking and confidence.

Field	Purpose
Rating	Best current estimate of player competitive strength.
Rating deviation	Uncertainty around the estimate; higher for new or inactive players.
Volatility	How rapidly performance appears to change.
Match count	Supports placement, confidence, and behavior analysis.
Human/bot split	Prevents bot-heavy records from being interpreted as equivalent to human competition.

Initial tuning values

The following values are accepted starting points, not sacred constants:

Parameter	Initial value
Initial rating	1500
Initial rating deviation	350
Initial volatility	0.06
Placement matches	5
Season duration	90 days

Performance profile beside Glicko

Glicko must not operate alone. Survival behavior is a direct, interpretable skill signal and should be used to prevent clearly unfair matches while ratings are uncertain or temporarily inaccurate.

- Average survival duration.
- Median survival duration.
- Average over the last 10 matches.
- Average over the last 25 matches.
- Personal and recent best survival duration.
- Survival-time variance.
- Tap frequency and tap efficiency.
- Recovery success rate.
- Item pickup and item-use effectiveness.
- Hazard avoidance.
- Disconnect and abandonment rate.

Why time is not the visible rating itself

A survival average is valuable, but it does not fully capture opponent quality, mode context, consistency, or strategic interaction. Glicko measures results against rated opponents; the performance profile explains how those results are being produced. Together they are stronger than either alone.

9. Matchmaking

Match quality principle

The target opponent should make the player think: “I could beat them if I perform slightly better than usual.” A 63-second player facing a 72-second player may be productively challenged. A 15-second player facing a 72-second player is not meaningfully competing.

Compatibility model

Factor	Initial weight	Purpose
Glicko similarity	70%	Primary estimate of competitive equality.
Survival-performance similarity	20%	Prevents obvious mismatch and catches rating lag.
Connection quality	10%	Protects responsiveness and result integrity.

Weights are tunable through analytics and should not be hard-coded into the client.

Preferred survival compatibility

Average survival	Preferred opponent range
0-15 seconds	Approximately within 8 seconds.
16-30 seconds	Approximately within 12 seconds.
31-50 seconds	Approximately within 15 seconds.
51-75 seconds	Approximately within 18 seconds.
76+ seconds	Use rating, percentile, and high-skill population data rather than a fixed interval.

Expanding search

Queue time	Behavior
0-5 seconds	Very close rating and performance; real players only.
5-10 seconds	Slightly expand rating, survival range, and region within latency tolerance.
10-15 seconds	Expand further while preserving a credible expected win range.
After threshold	Insert an appropriately rated bot when mode and tier rules allow.

The expected win probability should generally remain around 40%-60%, with widening allowed only after longer waits and never to the point of a clearly predetermined result.

New, returning, and rapidly improving players

- **New player.** High rating deviation; placement matches; performance safeguards; faster early adjustment.
- **Returning player.** Increased deviation after inactivity; previous rating retained as an anchor; short recalibration period.
- **Rapidly improving player.** Recent survival profile and win streak can expand upward search sooner, reducing smurf-like low-rank domination.
- **Inconsistent player.** Median, variance, and rating deviation prevent one extraordinary run from defining all future matches.

10. Bots and Ghost Opponents

Purpose

Bots solve the multiplayer cold-start problem and guarantee that a competitive player can enter a meaningful match even when concurrent human population is low. They are not a substitute for community; they are population insurance, practice opponents, onboarding tools, and casual backfill.

Honesty requirement

PLAYER TRUST Bots must be identified as CPU, AI Challenger, or another clear label. They may have names, heroes, cosmetics, badges, and personalities, but they must not impersonate human accounts.

Same-rule requirement

- Same tap limits and input rules.
- Same gravity and movement physics.
- Same momentum and collision behavior.
- Same character statistics and legal abilities.
- Same item information available through perception.
- No future-spawn knowledge.
- No impossible reaction speed.
- No guaranteed recovery.

Human imperfection model

Bot difficulty should come from decision quality, timing consistency, awareness, and recovery—not hidden power. Bots should occasionally tap early or late, overcorrect, under-correct, panic near the ground, misjudge momentum, miss or ignore items, overcommit to value, and fail recoveries. Error timing must vary so bots do not reveal deterministic patterns.

Bot personality archetypes

Archetype	Behavior
Floater	Conservative, altitude-focused, low item risk, consistent survival.
Aggressor	Pursues disruptive opportunities and risky items; creates pressure but overcommits more often.
Climber	Seeks maximum altitude and powerful upward movement; may struggle with forced downward states.
Technician	Efficient taps, strong recovery, low variance, calculated item use.
Trickster	Unpredictable movement and unusual priorities; difficult to read.
Panicker	Performs adequately until near-ground pressure causes poorer decisions.

Item recognition and reaction

Current bot behavior has felt too reactive and too quick when collecting items. The bot pipeline should explicitly separate perception, recognition, evaluation, commitment, movement, and collection.

Bot skill	Recognition + decision delay	Behavior
Beginner	600-1,000 ms	Small awareness, frequent misses, poor prioritization.
Developing	450-800 ms	Recognizes obvious items, hesitates, chases inefficiently.
Intermediate	300-650 ms	Reasonable awareness and mixed risk judgment.
Skilled	200-500 ms	Fast but fallible evaluation and route adjustment.
Advanced	140-350 ms	Strong awareness with occasional hesitation or bad pursuit.
Elite	100-280 ms	Very fast but still randomized and non-perfect.

Item pursuit probability

Bot level	Initial useful-item pursuit range
Beginner	25%-45% of recognized useful items.
Intermediate	45%-70%.
Skilled	65%-85%.
Elite	75%-92%, still dependent on survival risk and personality.

Bot performance targeting

A bot should have a performance distribution rather than an exact programmed death time. For a player averaging 63 seconds, an appropriate bot might average 60-68 seconds, commonly finish in 52-72 seconds, occasionally fail around 40-50 seconds, and occasionally reach 73-82 seconds. This produces rivalry rather than a disguised stopwatch.

Ghost opponents

Ghosts replay recorded player input, movement, ability use, and performance. They should be labeled as recorded challenges. Ghosts are especially suitable for daily seeds, time trials, biome races, asynchronous friend challenges, and low-population competition where live reaction is unnecessary.

11. Seasons, Tiers, and Rewards

Visible ladder

Tier	Divisions	Progression format
Bronze	V, IV, III, II, I	3 stars per division; floor at Bronze V.
Silver	V, IV, III, II, I	4 stars per division; floor at Silver V.
Gold	V, IV, III, II, I	5 stars per division; floor at Gold V.
Platinum	V, IV, III, II, I	5 stars per division; floor at Platinum V.
Diamond	V, IV, III, II, I	5 stars per division; floor at Diamond V.
Master	No standard divisions	Rating/leaderboard placement; human competition prioritized.
Grandmaster	Leaderboard rank	Highest prestige tier; human-only competitive results.

Hearthstone-inspired progression

- A normal win grants one visible rank star.
- A normal loss removes one star where demotion rules permit.
- A win streak begins after three consecutive wins and may grant two stars per win in lower/middle ranks.
- Win-streak bonuses stop at Diamond or Master to protect upper-ladder integrity.
- Glicko updates remain mathematically independent of visible star bonuses.
- Major tier floors prevent falling into a lower tier after reaching Bronze, Silver, Gold, Platinum, Diamond, or Master floors during that season.

Season cadence

CURRENT CANON Competitive seasons last 90 days. Visible progression resets each season, while historical accomplishments remain permanent and Glicko undergoes a soft compression rather than a complete wipe.

Proposed soft reset formula:

$$\text{New rating} = 1500 + ((\text{old rating} - 1500) \times 0.60)$$

The compression factor should remain remotely configurable. Rating deviation should increase at season start so the system can reassess players more quickly.

Season reset goals

- Create a renewed climb without forcing elite players through the full beginner population.
- Provide clear ninety-day goals and content cadence.
- Make seasonal cosmetics and rank records meaningful.
- Prevent permanent leaderboard stagnation.
- Retain a player's lifetime identity and historical record.

Season rewards

Rewards are based primarily on the highest tier reached during the season, not only the final-day position. Rewards must remain cosmetic or presentation-focused.

Tier reached	Illustrative reward class
Bronze	Participation badge and Hero Coins.
Silver	Additional Hero Coins and profile badge.
Gold	Seasonal tap effect.
Platinum	Animated badge or profile frame.
Diamond	Exclusive character particle effect.
Master	Exclusive seasonal character variant or equally prestigious cosmetic.
Grandmaster	Numbered leaderboard badge and premium prestige cosmetic.

Bots by tier

Tier	Bot policy
Bronze	Allowed; normal visible progression; rating gains may be normal or slightly reduced.
Silver	Allowed; transparent labeling; anti-farming monitoring.
Gold	Allowed with reduced frequency and possibly reduced rating gains.
Platinum	Rare; human preferred.
Diamond	Emergency backfill after extended wait; reduced gains and losses.
Master	Human only.
Grandmaster	Human only; leaderboard-critical.

12. Infinity Taps

Mode definition

CURRENT CANON Infinity Taps is a local, noncompetitive, no-loss mode designed for fidgeting, stress relief, exploration, discovery, and permanent tap accumulation. Coins do not spawn as pickups. Every 1,000 valid taps awards one Hero Coin.

Emotional goal

Infinity Taps should feel like a playful world that is happy the player is touching it. The player is not trying to avoid failure. They are moving, experimenting, listening, discovering, collecting, and seeing how the world responds. Its core question is not “Can I survive?” but “What happens if I tap this, go farther, or keep tapping?”

Movement

- The player begins on the ground rather than falling into immediate danger.
- Tap position moves the character left, right, upward, or downward depending on context and world physics.
- Movement should preserve the tactile identity of competitive TAM while allowing freer traversal.
- The world can extend vertically and horizontally through connected biomes, depths, atmosphere, space, moons, and planets.
- No movement mistake should permanently end the session; it may redirect, bounce, sink, launch, or reveal an event.

Tap economy

Rule	Requirement
Valid tap	Must be accepted by gameplay, not menu tapping, automation, or invalid spam.
Permanent count	Lifetime Infinity tap total persists across sessions and devices for authenticated users.
Hero Coin conversion	1 Hero Coin for every 1,000 valid Infinity taps.
No coin spawn	Infinity Taps does not scatter normal coin pickups; rewards come from taps, discoveries, milestones, and exclusive content.
Anti-automation	Server or trusted validation must prevent macros from becoming the optimal economy strategy.

Two-minute zone rhythm

Biomes or zones are intended to take approximately two minutes to traverse under normal active play. This length lets the player experience most of a musical idea, encounter several reactions, and feel progress without making every biome a long commitment. The exact duration may vary for secrets, branches, depth, boosts, or player behavior.

Infinity boosts and modifiers

Modifier	Intended behavior
Velocity Increase	Movement accelerates, changing the traversal rhythm and reaction window.
Tap-and-Hold Jet Engine	Holding input creates sustained thrust rather than individual impulses.
Anti-Gravity / Inverted Tap Phase	A special rule where taps force the player downward, reversing normal expectation and making descent intentional.
Extra Bouncy	The character launches absurdly high or rebounds dramatically from surfaces.

These modifiers are Infinity experiences, not paid power. They should be temporary discoveries, zone rules, events, or unlockable play variations.

Why Infinity rewards must be exclusive

Skins, effects, variants, badges, companions, and other rewards earned through Infinity Taps cannot be purchased with Hero Coins, money, the seasonal pass, supporter bundles, ads, or loot systems. They must communicate that the

player explored, persisted, or discovered something. This is critical because it gives noncompetitive players status that cannot be replicated by spending or ranked skill.

13. Infinity Taps World and Biomes

World journey

The mature Infinity Taps world can move from ordinary natural spaces into extreme environments and eventually the impossible. The transition should feel like a continuous adventure rather than a level-select checklist, while still permitting fast travel after discovery.

Broad region	Example biomes / layers	Distinctive movement or events
Grounded nature	Field, meadow, farmland, forest, jungle.	Animals, plants, wind, crops, insects, ponds, hidden paths.
Elevation	Hills, mountains, cliffs, high forest.	Updrafts, rock reactions, waterfalls, caves, falling debris.
Arid and extreme	Desert, dunes, volcanic fields, lava regions.	Sand bursts, heat shimmer, lava launches, obsidian reactions.
Frozen world	Arctic/Antarctic ice, glaciers, snow fields.	Sliding, cracking ice, snow accumulation, aurora, frozen creatures.
Water journey	Shore, surface ocean, reef, deep ocean, abyss, Mariana-Trench-like depth.	Splashes, currents, bubbles, whales, pressure ambience, bioluminescence.
Atmosphere	Cloud layer, storm layer, upper atmosphere.	Wind, rain, lightning, cloud bounce, reduced gravity.
Space	Orbit, meteor fields, satellites, Moon.	Low gravity, rotation, meteor impacts, moon dust.
Other worlds	Planets, alien landscapes, strange moons.	Alien wildlife, tractor beams, unusual physics, rare heroes.

Specific interaction concepts

- **Lava launch.** Tapping or contacting a lava feature may launch the character upward instead of acting as a death surface.
- **Whale blow.** A whale encounter can propel the character from deeper water toward the surface.
- **Ocean current.** A current may carry the player into another path, depth, or secret region.
- **Meteor spin.** A meteor can rotate or redirect the character through space.
- **Alien tractor beam.** A rare encounter may relocate the character into a ship, hidden area, or alternate route.

Discovery density

A biome cannot rely only on background art. Each two-minute zone needs a layered content budget: common reactions, repeated-tap reactions, at least one movement modifier, environmental audio, several decorative behaviors, possible animals, a discovery or milestone, and rare content. Empty distance undermines the mode's promise.

Rare discovery framework

A rare per-tap discovery chance as low as 0.001% was discussed for extraordinary events or cosmetics. Such probabilities must include pity protection, duplicate rules, eligibility checks, and clear analytics. Rarity should create stories, not endless invisible frustration. Some discoveries should be deterministic milestones or exploration achievements rather than pure random chance.

14. Living-World Interaction Design

Primary rule

LIVING WORLD RULE Nearly everything visible in Infinity Taps should acknowledge the player's taps. First taps, repeated taps, and sustained attention should often produce different reactions.

Reaction ladder

Stage	Purpose	Example
First acknowledgment	Confirms the object is interactive.	Bush wiggles; water splashes; rabbit looks toward the player.
Repeated response	Rewards curiosity and creates personality.	Leaves fall; rabbit hops away; flower opens; fish changes direction.
Escalated reaction	Creates comedy, surprise, or mechanical change.	Animal uncovers an item; snow forms a creature; object launches the hero.
Discovery / exhaustion	Prevents infinite identical animation and marks completion.	Journal entry, rare variant, object settles, new route opens.

Established environmental examples

- **Water.** A tap creates a splash or ripple; repeated taps can produce larger splashes, waves, bubbles, fish reactions, or a hidden event.
- **Bush.** A tap makes it wiggle; repeated taps may shake loose leaves, fruit, insects, a creature, or a hidden object.
- **Tree leaves.** Tapping causes leaves to fall; repeated tapping may change the intensity, reveal nests, or alter seasonal particles.
- **Tree trunk and branches.** Can shake, bend, knock loose objects, or trigger resident animals.
- **Grass and flowers.** Can bend, bloom, release seeds, change color, or attract insects.
- **Rocks and ground.** Can produce dust, cracks, pebbles, echoes, hidden passages, or geological reactions.

Animal examples

Animal	First response	Repeated response / potential discovery
Rabbit	Looks, twitches, or hops.	Runs, changes path, digs, or uncovers an item/secret.
Bird	Turns, chirps, or flaps.	Takes flight, drops a feather, joins a flock, or leads toward a point of interest.
Squirrel	Startles or shifts position.	Climbs, throws an object, reveals a stash, or reacts to tree tapping.
Frog	Croaks or hops.	Creates a rhythm, dives, causes ripples, or reveals a pond event.
Fish	Changes direction.	Schools, jumps, creates bubbles, or interacts with deeper-water creatures.

Repeated tapping personality

Repeated taps should not always be interpreted as aggression. Depending on the object, they may be playful, musical, curious, annoying, surprising, or affectionate. The goal is to make the world feel authored and responsive without training the player that relentless tapping always yields a material reward.

Discovery Journal

The Journal records encountered biomes, animals, heroes, variants, environmental reactions, rare events, altitude/depth milestones, and secrets. It should reveal enough missing structure to motivate exploration without spoiling every surprise. Journal completion can award Infinity-exclusive cosmetics and profile recognition.

15. Characters, Heroes, and Variants

Character role

Heroes are the player's visible identity across competitive and Infinity play. They should be readable at mobile scale, expressive under repeated motion, compatible with particles and skins, and distinct enough that variants remain recognizable as the same base character.

Known character direction: Velvet Hex

Velvet Hex is an established character concept: a pixelated 2D tuxedo-cat-like plush demon. The character combines cute plush proportions with mischievous demonic elements. A vampire/Dracula-inspired alternate variant preserves the tuxedo body and tie while adding a red, black, and purple-trimmed trench coat, red horns, and visible vampire teeth. The variant must still read immediately as Velvet Hex rather than a separate character.

Hero and variant rules

- Base hero identity remains visually recognizable across variants.
- Variants may change costume, theme, color, particles, idle behavior, entrance, or presentation, but not ranked physics or power.
- Owned heroes can emote, wave, or acknowledge the player when encountered in Infinity Taps.
- Unowned heroes can open a lightweight identification and acquisition interface without fully pausing exploration.
- An unowned variant must clearly show whether it is purchasable, season-limited, achievement-earned, or Infinity-exclusive.
- Infinity-exclusive variants cannot be purchased through any other channel.

In-world hero encounter card

Field	Display
Hero name	Base character identity.
Variant name	Specific appearance or theme.
Rarity/status	Collection classification without implying competitive power.
Ownership	Owned, unowned, or base owned/variant unowned.
Acquisition	Hero Coins, direct bundle, seasonal pass, achievement, or Infinity-only requirement.

Legacy terminology: Puffs and artifacts

Early exploratory discussion described cute collectible creatures called Puffs, collectible artifacts, an Explore/Journal/Profile structure, trading, guilds, and light MMO-style systems. The current airborne survival design uses "heroes" more consistently. Puffs may be reconciled as a species, companion category, early hero class, or retired terminology. Artifacts may become cosmetic discoveries, lore collectibles, or Infinity objects, but should not introduce equipment power into ranked play without a separate standardized ruleset.

16. Progression, Achievements, Badges, and Challenges

Progression philosophy

Short rounds need long-term meaning. Progression should recognize survival, skill, persistence, experimentation, collection, exploration, social participation, and unusual behavior. It must not require spending to complete core achievement sets.

Tracked progression dimensions

- Lifetime valid taps and Infinity taps.
- Best, average, median, and recent survival time.
- Distance, altitude, depth, zones, worlds, and biomes reached.
- Matches played, human wins, bot wins, losses, streaks, and close finishes.
- Glicko rating, visible tier, seasonal peak, and leaderboard position.
- Green Apples, Poison Fruits, Golden Hero Tokens, and Hero Coins collected or used.
- Heroes, variants, cosmetics, badges, journal entries, and secrets unlocked.

Achievement categories

Category	Examples of recognized behavior
Fruit	Green Apple collection, Poison Fruit consumption, surviving poison, balancing both fruit types.
Survival	First flight, time thresholds, close recoveries, high gravity, long sessions.
Tapping	Lifetime taps, efficient taps, daily consistency, Infinity milestones.
Coins and Shop	Holding Hero Coins, earning tokens, first purchase, collection breadth.
Multiplayer	First victory, streaks, comeback, photo finish, friend play, crowded matches.
Ranked	Tier milestones, season completion, peak rating, Grandmaster placement.
Infinity Taps	Biome discovery, altitude/depth, animals, environmental reactions, rare events, journal completion.
Secret	Hidden until completed; rewards strange or risky behavior.
Seasonal	Limited ninety-day objectives and season-specific accomplishments.

Established achievement examples

Achievement	Requirement	Reward / identity
Life of Poisonous Fruit	Consume 500 Poison Fruits.	10 Hero Coins + Poison Fruit Badge.
Anti-Gravity	Collect/consume 500 Green Apple fruits.	10 Hero Coins + special Golden Apple cosmetic/badge.
Coin Collector	Hold 500 Hero Coins.	25 Hero Coins + Coin Collector Badge.
First Flight	Survive 30 seconds.	5 Hero Coins + Tiny Wings-style badge.
First Victory	Win the first online match.	10 Hero Coins + First Crown Badge.
Completionist	Complete all standard achievements.	Exclusive skin + Completionist Badge.

Tiered achievement example

Tier	Poison Fruit threshold	Reward
Bronze	25	3 Hero Coins.
Silver	100	5 Hero Coins.
Gold	500	10 Hero Coins + Poison Fruit Badge.

Tier	Poison Fruit threshold	Reward
Mythic	2,500	50 Hero Coins + animated Poison Badge.

Known achievement and badge vocabulary

The design and current asset/catalog work have used names including: Achievement Hunter, Apple Basket, Apple Calendar, Badge Case, Balance, Balance Fruit, Beginner, Champion Wings, Coin Collector, Comeback, Completionist, Crowded Sky, Dropped Coin, Empty Sky, Eternal Wings, Falling Star, First Crown, Floor, Friends, Fruit Salad, Golden Apple, Golden Hero, Golden Vault, Gravity, Greedy Hero, Last Tap, Merchant, Natural Flyer, Never Quit, Nine Lives, Perfect Crown, Photo Finish, poison tier badges, Poison Shield, and related progression variants. Exact thresholds and catalog mapping should be treated as implementation data rather than hard-coded UI copy.

Secret achievements

- **Was That Worth It?** Reward an intentionally questionable risk or low-value pursuit.
- **Floor Inspector.** Recognize repeated or unusual contact with the ground.
- **Maximum Poison.** Survive or act under an extreme poison/gravity condition.
- **Perfectly Balanced.** Meet a precise balance condition involving opposing fruit effects or outcomes.
- **Velvet's Nine Lives.** Perform a character-specific sequence of recoveries or survivals.
- **Too Greedy.** Lose or nearly lose while pursuing excessive value.

Badge display rules

- Badges are permanent profile collectibles and are not consumed when equipped.
- One primary badge appears beside the username.
- Up to three additional showcase badges appear on the profile.
- Badges can appear in multiplayer introductions, match results, leaderboards, and profile views.
- The server validates that all displayed badges are owned.
- Tiered achievements display the highest earned version rather than cluttering the profile with every lower tier.
- Secret achievements remain hidden or cryptic until unlocked.

Daily challenges

- Collect 10 Green Apples.
- Survive for 60 seconds.
- Play five rounds.
- Collect three Golden Hero Tokens.
- Recover from two close calls.
- Complete a round without collecting fruit.

Weekly challenges

- Accumulate 20 minutes of survival time.
- Win 10 online matches.
- Collect 100 Green Apples.
- Consume 25 Poison Fruits and survive.
- Earn 100 Hero Coins.
- Complete five daily challenges.

Challenge timing should use a consistent server-defined schedule, previously scoped around UTC, while the UI clearly shows the player's local expiration time.

17. Currencies, Economy, Shop, and Monetization

Economy principles

- Hero Coins are a collection and cosmetic currency, never a source of ranked power.
- Paid offers must be understandable and direct; avoid manipulative obscurity.
- Ads must not interrupt live competitive rounds.
- Infinity-exclusive rewards are excluded from every purchase path.
- Season-exclusive rewards remain earnable only during their season unless an explicit archival policy is created later.
- Every transaction and reward should be server-authoritative and idempotent.

Established offers

Offer	Price	Contents / rule
Hero Coin Starter Package	\$0.99	Entry-level Hero Coin purchase; exact coin amount anchored around the initial economy.
30-Day Ad-Free Pass	\$2.99	Removes ads for 30 days, not permanently.
Seasonal Pass	\$4.99	One 90-day season's cosmetic reward track; unique season-only cosmetics.
Supporter Bundle	\$9.99	Character bundle + character particle effect + tap effect + 60 days ad-free.

Hero Coin pack framework

Price	Proposed Hero Coins
\$0.99	100
\$4.99	600
\$9.99	1,300
\$24.99	3,500
\$99.99	15,000

These pack sizes are an accepted economy anchor but remain subject to store policy, regional pricing, conversion analysis, and cosmetic catalog balancing.

Cosmetic price bands

Category	Proposed range
Tap effect	100-400 Hero Coins.
Character particle effect	200-600 Hero Coins.
Character skin / variant	400-1,200 Hero Coins, unless direct/seasonal/Infinity-exclusive.

No-pay-to-win enforcement

ABSOLUTE RESTRICTION Paid content must never alter jump force, gravity, speed, collision, tap acceptance, item spawn rate, Hero Coin earnings, rare-discovery chance, achievement speed, matchmaking, Glicko rating, ranked rewards, or any competitive advantage.

Ad policy

Ads may support free access, optional rewards, or out-of-round monetization, but they must not interrupt active matches. The 30-day ad-free pass is intentionally renewable rather than permanent. Any rewarded-ad system must not make watching ads the dominant progression strategy or bypass Infinity-exclusive content.

18. Cosmetics, Tap Effects, and Character Particles

Cosmetic layers

Layer	Examples
Hero appearance	Base heroes, skins, themed variants, color treatments.
Character particles	Continuous or movement-linked trails and auras.
Tap effects	Visual/audio reaction at the player's tap location.
Movement and landing effects	Trails, splashes, dust, space wake, impact reactions.
Presentation	Spawn/entrance animations, victory animations, profile frames, backgrounds.
Identity	Badges, titles, emotes, companion creatures, ambient themes.

Equip rules

- One primary character particle effect is equipped at a time.
- Tap effects and character particles are separate slots and can be combined creatively.
- Cosmetics must preserve character, item, hazard, and opponent visibility.
- Competitive modes may reduce or standardize extreme visual intensity while preserving ownership and theme.
- Owned cosmetics are server-validated before display.

Established tap effects and pricing

Tap effect	Price	Behavior
Purple Pulsing Rings	100	Default colored rings pulse outward from each tap.
Additional Custom Color	50	Color picker supports hue/RGB and darkness or value adjustments.
RGB Pulsing Rings	300	Color changes randomly or cycles with each pulse/tap.
Rainstorm	200	Rain falls at tap location; approximately 0.001% lightning chance can scorch the ground.
Snowfall	200	Snow falls and briefly accumulates; a large pile may produce an evil snowman.
Magma / Lava	200	Molten reaction at tap location; may form or summon an obsidian golem.
Fire	75	Localized flame burst or temporary burning visual.
Autumn Leaves	200	Leaves spiral or fall from the tap location.

Expanded tap-effect catalog ideas

Approved ideation pool:

Bubble Pop; Pixel Lightning; Starfall; Galaxy Portal; Ghost Hands; Slime Splash; Ink Blot; Paint Burst; Flower Bloom; Mushroom Growth; Crystal Growth; Mini Earthquake; Water Splash; Ocean Wave; Sand Burst; Candy Explosion; Confetti Cannon; Musical Notes; Retro Arcade Hit; Glitch Effect; Shadow Creatures; Tiny Footprints; Magic Runes; Black Hole; Cloud Puff; Fireworks; Miniature Meteor; Hearts and Affection; Bat Swarm; Tiny Companion Reaction.

The strongest early additions identified were Bubble Pop, Slime Splash, Musical Notes, Flower Bloom, Retro Arcade Hit, Magic Runes, Galaxy Portal, Tiny Footprints, Confetti Cannon, and Glitch Effect.

Character particle catalog ideas

Purple Slime Droplets; Shadow Smoke; Golden Sparkles; Poison Bubbles; Pixel Stars; Fire Trail; Blue Ghost Fire; Snowflake Trail; Autumn Leaf Trail; Flower Petals; Bubble Trail; Ocean Foam; Lightning Static; Rainbow Trail; RGB Pixel Trail; Glitch Fragments; Cosmic Dust; Mini Comets; Black Hole Orbit; Tiny Bats; Ghost Companions; Tiny Slimes; Cat Paw Prints; Demon Footprints; Hearts; Musical Notes; Confetti; Candy Trail; Crystal Shards; Mushroom Spores; Fireflies; Cloud Trail; Rain Cloud Companion; Halo Light; Dark Aura; Crown Sparkles; Retro Score Numbers; CRT Afterimage; Matrix Code; Floating Runes; Moon Dust; Alien Spores.

Infinity-exclusive cosmetic classes

- Character skins and variants.
- Tap effects and particle trails.
- Landing, movement, water, and space effects.
- Profile badges and frames.
- Character colors.
- Companion creatures.
- Ambient sound themes.
- Emotes and idle animations.
- Milestone auras and discovery-linked reactions.

Example names previously proposed include Meadow Velvet Skin, Abyssal Velvet Skin, Moon Dust Tap Effect, Endless Ripple Effect, and Million Tap Aura. These names are concept-level until catalog approval.

19. Audio and Music Direction

Overall audio identity

TAM audio should feel bright, tactile, playful, slightly mischievous, and immediately reactive. The soundscape combines electronic rhythm with small organic sounds so the game feels alive rather than mechanically sterile.

Infinity Taps main loop

- Bright electronic percussion.
- Bouncy synth bass.
- Plucked marimba, kalimba, or pizzicato-style melodies.
- Soft arcade-inspired tones.
- Handclaps, snaps, wooden knocks, bells, whistles, and environmental ambience.
- Occasional unexpected musical reactions that make the world feel responsive.

The loop should be endlessly replayable, energetic without becoming stressful, and able to absorb biome layers and tap-triggered surprises.

Main menu music

The menu theme should be calmer, cleaner, welcoming, and lightly mischievous. It should introduce the same musical identity through softer percussion, warm bouncy synth bass, gentle marimba or pizzicato, subtle arcade tones, snaps, bells, wood taps, and restrained whistles. It should feel like the world is waiting to be touched.

Solo and competitive run music

Competitive music should be faster, sharper, and more focused than Infinity Taps. Use punchy electronic percussion, driving synth bass, quick plucked melodies, arcade tones, claps, snaps, whistles, and wood hits. Intensity should rise as gravity, survival time, opponent pressure, or a close finish escalates. The music should motivate precise action rather than become aggressive noise.

Biome-specific adaptive music

Infinity Taps should have biome-specific music while retaining a recognizable core theme. Do not store a separate full-length two-minute song for every biome. Use a shared foundation—rhythm, bass, motif, or harmonic identity—combined with compact biome melody, ambience, percussion, and reaction layers. This makes transitions coherent and reduces storage.

Reactive audio

- Valid taps trigger concise sounds matched to the equipped effect.
- Repeated taps can form rhythms or escalate environmental layers.
- Animals and objects respond with localized sound rather than generic global effects.
- Rare discoveries receive distinct signatures that remain recognizable later.
- Competitive state escalation can add layers rather than abruptly replace the track.
- Players can independently control music, ambience, tap sounds, character sounds, cosmetic effect sounds, and haptics.

Storage and compression direction

CURRENT AUDIO IMPLEMENTATION TARGET Preserve WAV or FLAC masters. Export runtime music primarily as Ogg Opus stereo at 48 kHz VBR: approximately 80-96 kbps for calmer tracks and 96-112 kbps for competitive tracks. Stream long tracks or use compressed-in-memory playback; do not decompress all music on load.

- Remove unnecessary silence while preserving exact loop points.
- Use a shared import preset or build script to keep quality consistent.
- Use modular biome packages targeted around 250-600 KB where possible.
- Initial Infinity soundtrack target: approximately 5-8 MB rather than one full song per biome.
- The current eight-song set was approximately 27 MB; an initial optimized target discussed was roughly 10-15 MB total without perceptible quality loss.

20. Visual, UI, and Presentation Direction

Visual identity

The game is a cute, readable 2D mobile experience with enough personality to support strange, mischievous, supernatural, natural, and cosmic content. Pixel-art character work has been explored through Velvet Hex, but the critical requirement is consistency at portrait mobile scale rather than commitment to a single rendering technique in this document.

Portrait gameplay layout

- The active play space dominates the vertical screen.
- Critical survival information stays readable without obscuring the character or ground.
- Opponent information is compact and symmetrical where possible.
- Tap locations, effects, and pickups remain visible under thumbs.
- Menu and loadout controls use large mobile touch targets.
- Accessibility options should reduce flash, particles, shake, haptics, and sound categories independently.

Primary navigation

Area	Purpose
Play	Solo, local, private online, casual, ranked, Infinity Taps.
Heroes / Collection	Owned heroes, variants, skins, particles, tap effects, animations, companions.

Area	Purpose
Journal	Infinity biomes, creatures, reactions, discoveries, secrets, and collection progress.
Challenges / Achievements	Daily, weekly, permanent, secret, and seasonal goals.
Ranked	Tier, division, stars, season timer, peak, match history, leaderboard.
Shop	Hero Coin cosmetics, direct bundles, season pass, ad-free pass.
Profile	Primary badge, showcase badges, records, heroes, seasonal history, privacy settings.
Settings	Audio categories, haptics, graphics, accessibility, account, privacy, controls.

Profile presentation

Profiles should make identity legible without becoming a spreadsheet. The first view should show hero/variant, username, primary badge, visible rank, selected frame, core records, and three showcase badges. Deeper tabs can show match history, achievements, seasonal history, Infinity Journal, collection, and statistics.

Cosmetic clarity controls

Players should be able to reduce opponent cosmetic intensity, screen shake, flashes, particles, or sound without removing ownership. Competitive accessibility settings may simplify effects locally while still showing a recognizable icon, color, or compact version.

21. Social, Community, and Retention Systems

Current social foundation

- Accounts and cloud saves.
- Usernames and online profiles.
- Friend or opponent search.
- Private online matches.
- Public matchmaking.
- Leaderboards.
- Badge and cosmetic identity.

Future social layers

System	Purpose	Scope caution
Friends	Reliable rematches, private challenges, friend leaderboards.	High value; should precede broad social complexity.
Global and regional rankings	Prestige and comparison.	Requires anti-cheat and population context.
Shared world events	Community tap or discovery milestones.	Must remain achievable across population sizes.
Guilds / groups	Persistent social identity and cooperative goals.	Legacy-approved concept; post-MVP due moderation and feature load.
Trading	Exchange artifacts/items or collectibles.	High economy and fraud risk; not appropriate for early cosmetic economy.
Tournaments	Scheduled competitive prestige.	Requires stable human population and spectator/admin systems.

Retention without coercion

Retention should come from improving skill, completing achievable goals, collecting expressive rewards, discovering reactions, and anticipating ninety-day seasons. Avoid designing around fear of missing basic functionality, punitive streak loss, or purchased competitive rescue.

22. Technical Architecture and Current Implementation

Current stack snapshot

Layer	Current direction / implementation
Game client	Godot project with gameplay, UI, bot, network, achievements, catalog, stats, save, and authentication scripts.
Backend	Cloudflare Worker service.
Real-time rooms	Lobby and MatchRoom Durable Objects / WebSocket-oriented flow.
Database	Cloudflare D1 database named tap-against-me.
Authentication	Supabase authentication with verified-email handling and modern JWT/JWKS integration work.
Deployment	Worker deployed at tap-against-me.dontwatchmedia.workers.dev.
Testing	A validated backend test suite previously reported 53 passing tests and zero failures.

Known client modules

The project includes or has included modules for analytics, authentication, badge management, bot behavior, catalog, challenges, event bus, game configuration, main flow, match configuration, menus, network match/service, online match, pickups, player logic, player statistics, save service, statistics panel, Supabase authentication, achievement testing, and UI.

Known backend responsibilities

- Authentication and verified-user handling.
- Username/profile validation.
- Lobby and room management.
- Private rooms and matchmaking.
- WebSocket communication.
- Authoritative simulation and result validation.
- D1 persistence.
- Achievements, badges, rewards, matches, and flags.
- Leaderboards and profile privacy.
- Anti-cheat validation and server-side catalog ownership.

Implementation principles

- **Server authority.** Clients submit input and intent; they do not dictate ratings, rewards, ownership, or final match outcomes.
- **Deterministic rules where practical.** Shared simulation logic reduces client/server disagreement.
- **Remote configuration.** Matchmaking ranges, season dates, rewards, bot delays, and balance values should change without a full client release.
- **Idempotent rewards.** A match, achievement, purchase, or tap milestone cannot pay twice because of retries.
- **Versioned match rules.** Recorded results identify the simulation and content version used.
- **Secure production routes.** Schema bootstrap and administrative operations remain unavailable publicly after initialization.

Current milestone direction

The nearest technical proof is a complete two-device flow: real account creation/login, Google authentication, distinct player accounts, opponent discovery, lobby creation, synchronized match, authoritative result, persistence, and

visible profile/ranking update. Finishing this vertical slice protects the networking investment before expanding content scope.

23. Analytics, Integrity, Safety, and Anti-Cheat

Core analytics events

- Session start/end and mode selected.
- Run start/end, survival time, gravity phase, death reason.
- Tap count, accepted/rejected taps, tap timing distribution, movement efficiency.
- Pickup seen, pursued, collected, ignored, and effect outcome.
- Matchmaking wait, expansion stage, opponent type, expected win probability.
- Bot profile, reaction parameters, performance, and player sentiment proxies.
- Rating before/after, rank stars, promotion/demotion.
- Challenge progress, achievement completion, reward claims.
- Infinity biome time, interactions, discoveries, rare rolls, journal completion.
- Shop impressions, purchases, ad events, refunds, and entitlement failures.
- Disconnect, reconnect, abandonment, desync, and server validation flags.

Anti-cheat priorities

Threat	Mitigation
Automated tapping / macros	Input-rate analysis, impossible timing patterns, valid-tap rules, server validation, progression throttles where justified.
Client-modified physics	Authoritative simulation and versioned configuration.
Fake match results	Server-owned match completion and signed/validated state.
Reward duplication	Idempotent transaction IDs and unique match/client constraints.
Cosmetic spoofing	Server-validated ownership and loadout filtering.
Bot farming	Vary bot profiles, reduce repeated rewards, separate bot/human stats, require humans at high tiers.
Leaderboard abuse	Verified ranked matches only, anomaly detection, review flags, and result rollback tools.

Disconnect handling

- Provide a reconnection grace period.
- Track exact disconnect and reconnect timestamps.
- Differentiate client closure, network loss, and server failure where possible.
- Repeated abandonment counts as a loss and may create queue penalties.
- Verified server failure should not create normal rating loss.
- Log final result reason and match state for dispute analysis.

Privacy and reporting

Profiles should expose only intended public statistics. Match reports, cheating flags, and moderation information remain private. The product needs report, block, and profile privacy controls before broad social systems such as guilds or trading are introduced.

24. MVP, Production Phases, and Roadmap

Phase 0: Core proof already underway

- Airborne tap physics and ground-loss condition.

- Solo and bot-capable game logic.
- Accounts, profiles, authentication, backend routes, D1 persistence.
- Lobby/room networking and authoritative simulation.
- Achievements, badges, catalog, statistics, and test infrastructure.

Phase 1: Competitive vertical slice

- Real signup, email confirmation, login, and Google authentication.
- Two distinct accounts on two phones.
- Opponent search or invite.
- Private online room.
- Countdown, synchronized match, authoritative result, and match persistence.
- Solo survival, local play, and a baseline bot.
- Green Apple, Poison Fruit, and Golden Hero Token.

Phase 2: Credible competitive MVP

- Casual matchmaking with expanding search and bot backfill.
- Glicko-2 rating and five placements.
- Bronze through Diamond visible tiers, stars, floors, and ninety-day season configuration.
- Basic match history and leaderboards.
- Bot reaction delays, errors, and item perception tuning.
- Hero loadout, one particle slot, one tap-effect slot, badges, and core shop.
- Daily/weekly challenges and permanent achievements.
- No-interruption ad policy and core purchase entitlements.

Phase 3: Infinity Taps MVP

- One polished biome or a small connected set of biomes.
- No-loss movement and persistent valid-tap count.
- 1 Hero Coin per 1,000 valid taps.
- Reactive water, bushes, trees, ground, and several objects.
- One or two animals with repeated-tap reactions.
- One hero encounter.
- Discovery Journal foundation.
- A few Infinity-only achievements.
- One hidden cosmetic or hero variant that cannot be purchased.
- Adaptive music foundation.

Phase 4: First public release hardening

- Onboarding and tutorial.
- Performance and battery optimization on target Android devices.
- Accessibility controls.
- Store compliance, privacy policy, crash/error reporting.
- Economy and entitlement testing.
- Cheat/anomaly review tools.
- Season 1 content and rewards.
- Community feedback and telemetry dashboards.

Post-MVP competitive expansion

- Master and Grandmaster human-only ladder.
- Ghost challenges and replay support.

- Friend leaderboards and private tournaments.
- Additional competitive modes after survival is stable.
- Spectating and tournament tools.
- Advanced smurf, bot-farming, and disconnect detection.

Post-MVP Infinity expansion

- Two-minute biome journey structure.
- Biome-specific modular music and ambience.
- More animals, heroes, variants, secrets, and environmental stories.
- Velocity, jet-engine, inverted-tap, and extra-bouncy modifiers.
- Ocean depth through abyssal regions.
- Atmosphere, space, Moon, planets, and alien events.
- Seasonal worlds and long-term exploration progression.

Long-term ecosystem

At maturity, TAM can support a shared account and universe where competitive mastery and Infinity discovery produce different forms of prestige. A player may use an Infinity-earned variant competitively, display a ranked seasonal badge while exploring, or unlock decorative Infinity encounters through competitive achievements. Cross-mode rewards must remain cosmetic and should encourage curiosity rather than force players into a mode they dislike.

25. Risks, Open Decisions, and Scope Guardrails

Primary risks

Risk	Why it matters	Guardrail
Two-games-in-one scope	Competitive networking and Infinity content require different teams and production rhythms.	Finish the competitive vertical slice; build Infinity in controlled phases; preserve separation in UX and backlog.
Multiplayer cold start	Empty queues damage perception.	Solo, local, private, bots, ghosts, and transparent queue expansion.
Bot unfairness	Perfect reactions feel dishonest.	Human delay, perception, error, same-rule physics, performance distributions.
Infinity emptiness	Large worlds without reactions feel unfinished.	Content-density budget per biome and staged expansion.
Cosmetic overload	Effects can obscure gameplay or inflate app size.	Intensity controls, ranked simplification, modular assets, compression budgets.
Economy distrust	Short ad-free period or rare items may feel exploitative.	Clear terms, fair pricing, no pay-to-win, guaranteed non-purchasable achievements.
Rank frustration	Season reset can feel like progress deletion.	Soft Glicko compression, rank floors, permanent history, highest-rank rewards.

Open decisions that need playtesting or product approval

- Exact tap impulse and directional curve.
- Gravity escalation curve and expected survival distributions.
- Whether competitive items are shared, mirrored, or independently spawned.
- Exact rule definitions for First to Zero, Most Taps, and Dual Victory.
- Maximum queue wait before bot backfill by mode and region.
- Whether Bronze/Silver bot wins receive full Glicko gain.
- Visible numerical rating versus tier-only presentation.
- Exact season pass structure and reward quantity.

- Whether Infinity Taps remains one mode, becomes a separately downloadable game, or eventually becomes a sister title.
- How Puffs and artifacts fit the current hero-based canon.
- Final art production style and pixel-art consistency requirements.
- Launch biome count based on content density and app-size budget.

Scope exclusion rules

- Do not add combat, crafting, dialogue trees, or full quest systems to Infinity before the reactive exploration loop proves itself.
- Do not add equipment stats or purchased hero power to ranked competition.
- Do not build guilds, trading, or tournaments before profiles, friends, matchmaking, reporting, and economy integrity are stable.
- Do not produce a separate full song for every two-minute biome.
- Do not make rare random rewards the only path to meaningful Infinity progression.
- Do not treat completed backend work as justification for unlimited multiplayer feature expansion; finish the nearest complete product.

26. Acceptance Criteria and Definition of Done

Core gameplay

- A new player understands the survival rule without external explanation.
- Tap response is immediate and predictable.
- Ground contact resolves consistently.
- Gravity escalation creates readable early, middle, and late phases.
- Items create decisions rather than random outcomes.

Competitive and networking

- Two authenticated players can find or invite each other and complete a synchronized match.
- Server validates the outcome and persists it once.
- Disconnect and reconnection rules are testable.
- Match history, achievements, and rewards update correctly.
- Private, casual, ranked, solo, and bot paths do not corrupt one another.

Ranking and matchmaking

- Glicko rating, deviation, and volatility update correctly.
- Five placements complete and visible rank is assigned.
- A 15-second player is not normally matched against a 72-second player.
- A 63-second player may reasonably face a 72-second player.
- Search expands gradually and records each stage.
- Season reset preserves history and applies soft compression.
- Rank floors, stars, streaks, promotion, and demotion behave as configured.

Bots

- Bots are visibly labeled.
- Bots obey the same physics and item information rules.
- Item response includes randomized recognition and decision delay.
- Bots sometimes miss, ignore, or mishandle items.
- Performance varies around a target distribution.
- High-tier bot restrictions and anti-farming rules function.

Infinity Taps

- There is no traditional loss state.
- Valid taps persist and award one Hero Coin per 1,000.
- No normal coins spawn.
- The environment reacts to first and repeated taps.
- At least one animal has a multi-stage response.
- At least one hero encounter works.
- An Infinity-exclusive reward cannot be purchased from any other system.
- Music and ambience transition without obvious interruption.

Monetization and cosmetics

- No purchase changes competitive physics, rating, rewards, or drop odds.
- Ads never interrupt active rounds.
- The 30-day and 60-day ad-free entitlements expire correctly.
- Cosmetics are ownership-validated.
- Visual-intensity settings preserve competitive readability.
- Infinity-exclusive and seasonal restrictions are enforced server-side.

Appendix A - Detailed Ranked Data Model

Player ranked profile

player_id
season_id
glicko_rating
glicko_deviation
glicko_volatility
visible_tier
visible_division
rank_stars
highest_tier_season
highest_division_season
highest_tier_lifetime
highest_rating_season
highest_rating_lifetime
wins
losses
bot_wins
bot_losses
human_wins
human_losses
win_streak
best_win_streak
placement_matches_completed
placement_matches_required
ranked_status
updated_at

Player performance profile

player_id
average_survival_ms
median_survival_ms
recent_10_average_ms
recent_25_average_ms
best_survival_ms
recent_best_survival_ms
survival_variance
average_taps_per_second
recovery_success_rate
item_pickup_rate
item_use_success_rate
hazard_avoidance_rate
disconnect_rate
abandonment_rate
updated_at

Ranked match record

match_id
season_id
player_one_id
player_two_id
player_two_is_bot
bot_profile_id

player_one_rating_before
 player_one_rating_after
 player_two_rating_before
 player_two_rating_after
 player_one_survival_ms
 player_two_survival_ms
 winner_id
 result_reason
 queue_duration_ms
 matchmaking_range
 connection_quality
 started_at
 ended_at
 validated

Bot profile

bot_profile_id
 display_name
 personality
 glicko_rating
 target_survival_min_ms
 target_survival_max_ms
 reaction_delay_min_ms
 reaction_delay_max_ms
 item_awareness_radius
 item_pursuit_probability
 tap_accuracy
 recovery_skill
 error_frequency
 risk_tolerance
 active
 updated_at

Season record

season_id
 season_name
 season_number
 start_date
 end_date
 status
 rating_compression_factor
 placement_match_count
 reward_configuration
 created_at

Remote configuration

Season dates; tier thresholds; stars per division; rank floors; placement count; queue timing; rating and survival expansion ranges; bot insertion timing; bot reaction and pursuit ranges; bot availability by tier; rating rewards against bots; soft-reset compression; streak rules; seasonal rewards; reconnect grace; queue penalties; item and gravity balance; cosmetic intensity limits.

Appendix B - Consolidated Content Catalogs

Mode catalog

Solo Survival; Local Competitive; Private Online Match; Casual Matchmaking; Ranked Matchmaking; Infinity Taps; Survival/Last Airborne; First to Zero; Most Taps; Dual Victory; Free-for-All; Tournament/Elimination; Ghost Challenge; Daily Survival Seed; Friend Benchmark; Community Milestone.

Biome catalog

Field; Meadow; Farmland; Forest; Jungle; Hills; Mountains; Cliffs; Caves; Desert; Dunes; Volcanic Field; Lava Region; Arctic; Antarctic; Glacier; Snow Field; Shore; Surface Ocean; Reef; Deep Ocean; Abyss; Mariana-Trench-like Depth; Cloud Layer; Storm Layer; Upper Atmosphere; Orbit; Meteor Field; Satellite Region; Moon; Other Planets; Strange Moons; Alien Landscapes.

Infinity modifier catalog

Velocity Increase; Tap-and-Hold Jet Engine; Anti-Gravity/Inverted Tap Phase; Extra Bouncy; lava launch; whale blow; ocean current; meteor spin; alien tractor beam.

Hero/cosmetic catalog seed

Velvet Hex; Vampire/Dracula Velvet Hex; Meadow Velvet; Abyssal Velvet; Moon Dust Tap Effect; Endless Ripple Effect; Million Tap Aura; purple rings; custom rings; RGB rings; Rainstorm; Snowfall; Magma; Fire; Autumn Leaves; expanded tap-effect and character-particle lists in Section 18.

Achievement catalog seed

Life of Poisonous Fruit; Anti-Gravity; Coin Collector; First Flight; First Victory; Completionist; Achievement Hunter; Apple Basket; Apple Calendar; Badge Case; Balance; Balance Fruit; Beginner; Champion Wings; Comeback; Crowded Sky; Dropped Coin; Empty Sky; Eternal Wings; Falling Star; First Crown; Floor Inspector; Friends; Fruit Salad; Golden Apple; Golden Hero; Golden Vault; Gravity; Greedy Hero; Last Tap; Merchant; Natural Flyer; Never Quit; Nine Lives; Perfect Crown; Photo Finish; poison tier badges; Poison Shield; Was That Worth It?; Maximum Poison; Perfectly Balanced; Velvet's Nine Lives; Too Greedy.

Appendix C - Legacy Concepts and Reconciliation Notes

Earlier broad concept

An earlier version of Tap Against Me was described as a portrait, mobile-first, effectively infinite tapping game where players tapped bubbles, targets, enemies, or resources; collected Puffs and artifacts; improved stats/equipment; explored; used a Journal and Profile; joined guilds; participated in shared events; traded artifacts; and engaged in light MMO-style community systems. These ideas predate the current airborne survival rule and should not be treated as automatically canonical.

Elements worth preserving

- **Puffs.** Could become a creature species, companion class, collectible inhabitants of Infinity, or a subset of heroes.
- **Artifacts.** Could become lore discoveries, rare environmental objects, cosmetic unlocks, or Journal collectibles rather than stat gear.
- **Explore / Journal / Profile.** Directly compatible with Infinity Taps and current profile design.
- **Global/friend rankings.** Compatible with competitive and asynchronous systems.
- **Shared events.** Compatible after the core account, analytics, and content pipeline is stable.
- **Guilds and trading.** Potential long-term systems, but expensive in moderation, fraud prevention, economy integrity, and scope.

Contradictions to resolve

Legacy idea	Conflict	Recommended reconciliation
Stat/equipment improvement	Can create pay-to-win or unequal ranked physics.	Keep competitive stats standardized; use cosmetics, collection, or separate non-ranked modifiers.
Bubble/target tapping as core	Current core is directional airborne survival.	Use bubbles/targets as optional items, events, challenges, or Infinity interactions.
MMO label	Creates expectations of persistent shared worlds and large social systems.	Describe TAM by its actual modes; use “light social progression” until systems justify stronger language.
Trading	Conflicts with exclusivity, rarity, and purchase integrity.	Do not implement until item classes, binding rules, fraud systems, and economy goals are explicitly designed.

Final canon statement

TAM IDENTITY Tap Against Me is a touch-first airborne survival and competitive game with fair ranked progression, believable bot support, expressive cosmetics, permanent achievements, and a no-loss Infinity Taps world where tapping creates movement, reactions, discovery, and exclusive prestige.